

Inductive reactance (X_L). In Fig. 4, inductive reactance (X_L) linearly increases as frequency increases. At low frequency, inductive reactance is low and at high frequency, it is high. Inductive reactance is directly proportional to frequency.

In Fig. 5, inductive reactance linearly increases as frequency increases. Also, where resonance frequency is 649.7Hz, inductive reactance X_L is 81.6 ohms.

Capacitive reactance (X_C). The ideal case of capacitive reactance versus frequency is shown in Fig. 6. In this figure, capacitive reactance (X_C) gradually decreases as frequency in-

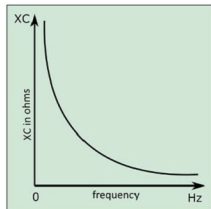


Fig. 6: Ideal case of capacitive reactance versus frequency

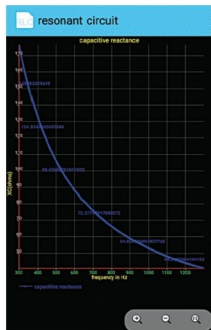


Fig. 7: Capacitive reactance versus frequency plot in Android

creases. For low frequency, capacitive reactance is high, and for high frequency, it is low. Capacitive reactance is inversely proportional to frequency. The capacitive reactance (X_C) plot obtained from the Android app for the example is shown in Fig. 7.

In Figs 7 and 8, capacitive reactance X_C decreases as frequency increases. Where resonance frequency is 649.74Hz, capacitive reactance X_C is 81.64 ohms.

Reactance (X). In Fig. 9, at resonance frequency, inductive reactance X_L and capacitive reactance X_C are equal. In other words, X_L and X_C are equal and opposite, so these cancel each other out. Therefore reactance is absent in the RLC circuit at resonant



Fig. 8: Capacitive reactance versus frequency plot in Android after zoom in

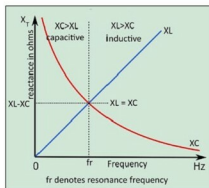


Fig. 9: Reactance versus frequency

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